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OVERVOLTAGE And UNDERVOLTAGE PROTECTION For Electrical Appliances

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In some places, especially in developing countries like ours, sudden ups and downs in the mains voltage are quite common. This sometimes causes a serious failure of electrical appliances like television sets, air-conditioners, and refrigerators, which are not designed to tolerate lower or higher voltages beyond a small limit.

This circuit incorporates a low-voltage and high-voltage tripping mechanism to protect the appliances from any damage due to voltage fluctuation. The author's prototype on breadboard is shown in Fig. 1. The circuit diagram is shown in Fig. 2.

The circuit's dual power supply using transformer X1 and diodes D1 through D4 isolates the supply between relay RL1 and control circuitry with the help of optocoupler IC PC817. It helps to eliminate the voltage-dropping problem when the relay energises. The capacitors C1, C2, and C3 are used to smoothen the rectified pulsating DC from the diodes.

The ICs IC1 and IC2 (TL431) are adjustable shunt regulators. The R1-R2 resistor voltage divider network is used to set the upper cut-off voltage and preset VR1 is used to set the lower cut-off voltage. Diode D5 works as a freewheeling diode and transistor T1 is used to drive

PARTS LIST

Semiconductors:	
IC1, IC2	- TL431 shunt regulator
IC3	- PC817 optocoupler
D1-D5	- 1N4007 rectifier diode
T1	- BC337 npn transistor
Resistors (all 1/4-watt, ±5% carbon), unless stated otherwise:	
R1	- 9-kilo-ohm
R2	- 1.2-kilo-ohm
R3	- 1-kilo-ohm
R4	- 4.7-kilo-ohm, 0.5W
R5	- 10-kilo-ohm
R6	- 1-kilo-ohm
VR1	- 10-kilo-ohm preset
Capacitors:	
C1, C2	- 100µF, 35V electrolytic
C3	- 1000µF, 35V electrolytic
Miscellaneous:	
X1	- 230V AC primary to 12V-0-12V, 500mA secondary
RL1	- 12V single changeover relay
CON1, CON2	- 2-pin terminal connector
	- 3-pin terminal connector

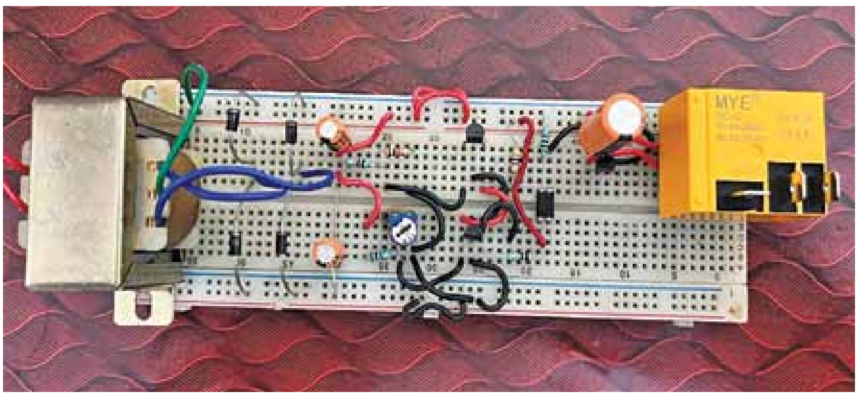

Fig. 1: Author's prototype on breadboard

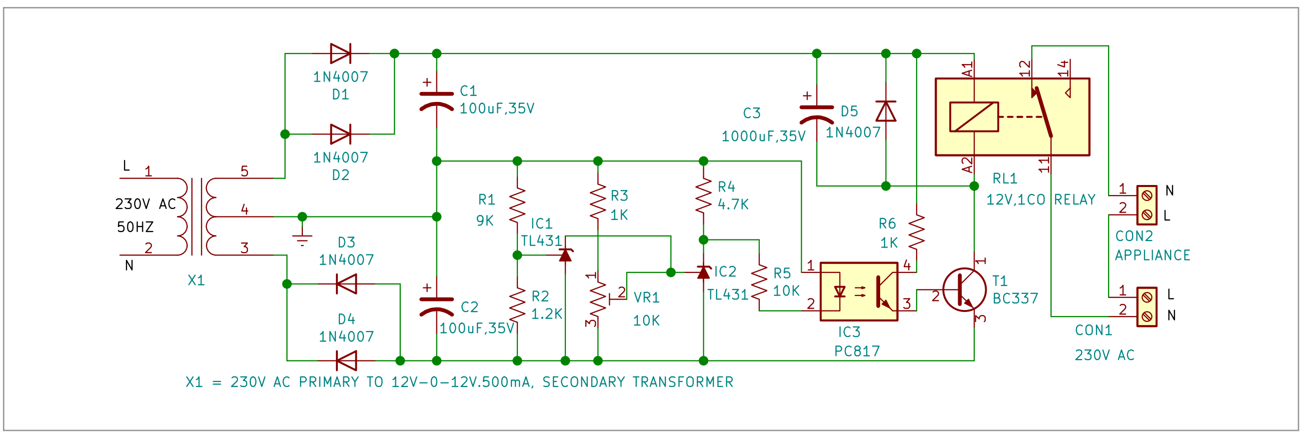

Fig. 2: Circuit diagram